



AeroTrade XAI: Autonomous Supply Chain Orchestrator

Explainable AI for high-stakes logistics decisions.

THE CHALLENGE

The Black Box Crisis in Global Logistics



Fragile Networks

Global supply chains are disruption-prone and increasingly impossible to manage manually at scale.



Complex Trade-Offs

Cost, time, risk, and carbon trade-offs exceed human cognitive bandwidth in real time.



No Trust, No Adoption

AI recommendations get rejected because managers cannot see the reasoning behind them.

An Agent That Shows Its Work



AeroTrade XAI is an autonomous agent built on **LangChain** that evaluates alternate shipping routes across four weighted factors – and exposes every calculation on a fully transparent dashboard.

Cost

Base + ID Variance × FX

Time

Base + ID Variance

Risk

Base + Weather Penalty

Carbon

Base + ID Variance

Built With Modern, Production-Ready Tools



Python

Core orchestration logic and agent runtime



Streamlit

Interactive dashboard UI and live controls



LangChain

ReAct agent loop and LLM integration layer



Qwen / GPT-4o

Large language model for reasoning and routing

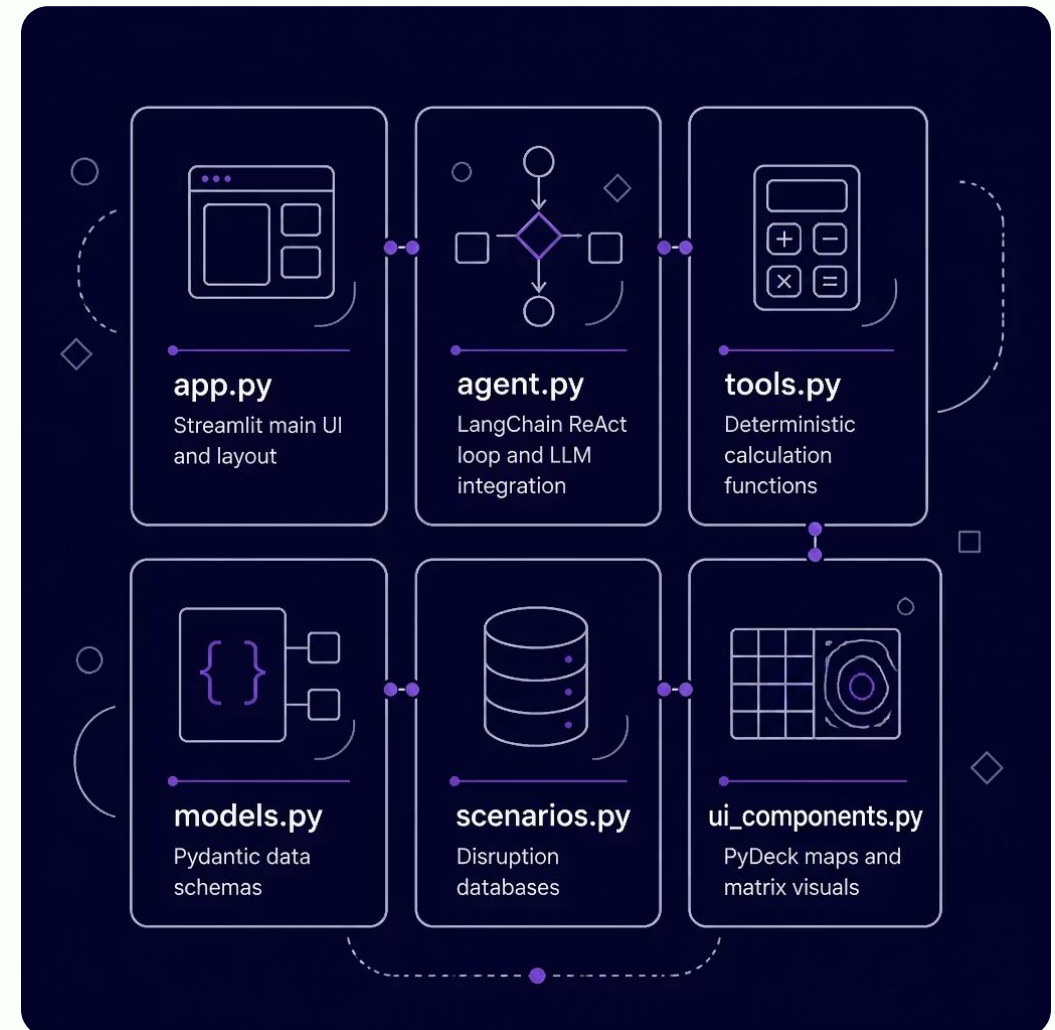


Pydantic

Strict data validation and schema enforcement

Six Modules, One Cohesive System

Each module has a single responsibility – from UI rendering to deterministic calculation – enabling clean testing, auditability, and extension.



No Guesswork — Just Auditable Formulas

Every score is computed from explicit, inspectable formulas – not opaque model weights. Managers can trace every number back to its source.

Cost Score

$$\text{Cost} = (\text{Base Cost} + \text{ID Variance}) \times \text{FX Rate}$$

Time Score

$$\text{Time} = \text{Base Time} + \text{ID Variance}$$

Risk Score

$$\text{Risk} = \text{Base Risk} + \text{ID Variance} + \text{Weather Penalty}$$

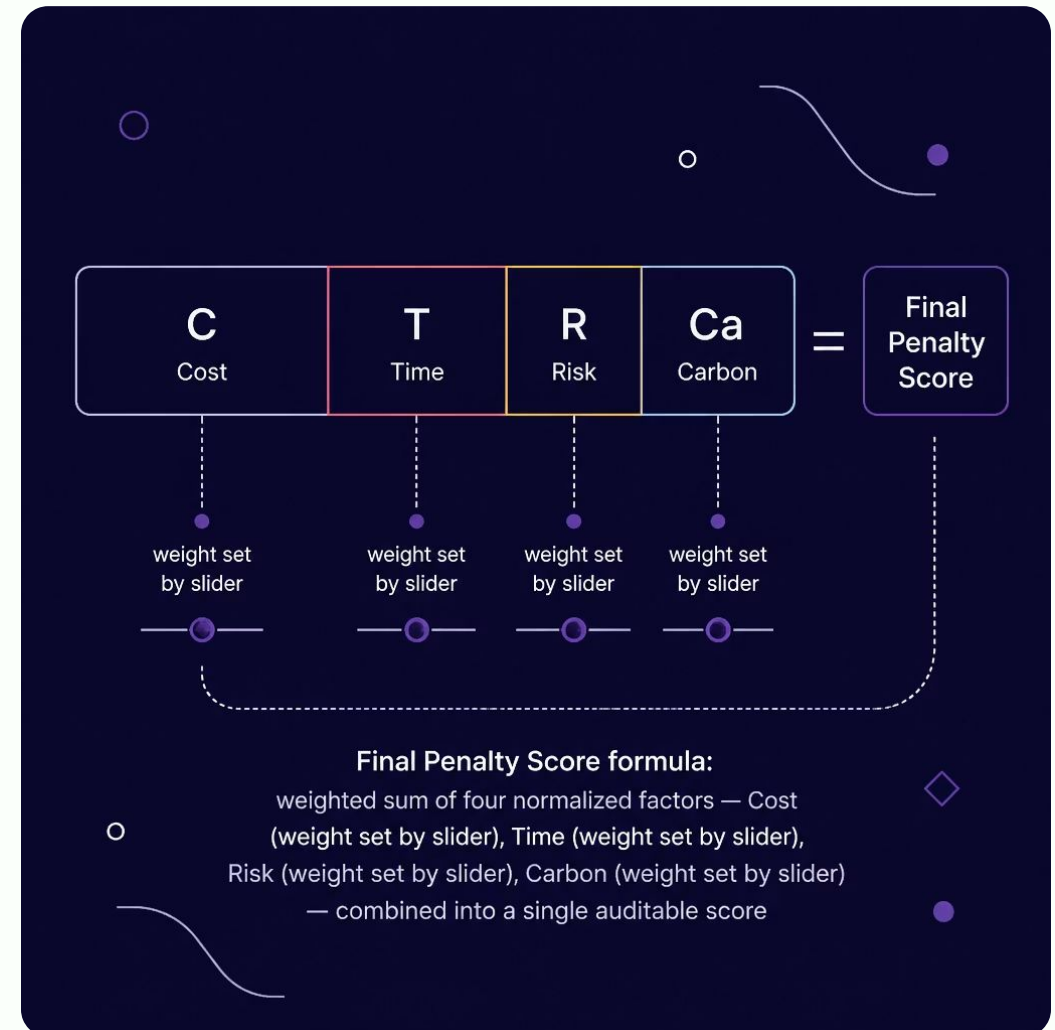
Carbon Score

$$\text{Carbon} = \text{Base Emissions} + \text{ID Variance}$$

User-Controlled Weighting

The **Final Penalty Score** is a weighted sum of normalized Cost, Time, Risk, and Carbon. Managers set weights live via UI sliders – giving direct control over trade-off priorities for every decision context.

- ❏ Weights sum to 1.0. Adjusting them instantly re-ranks all route alternatives.



What's Next for AeroTrade XAI



Real-Time API Integration

Live freight rate and capacity feeds via Flexport API



Live Disruption Feeds

Real-time weather, port congestion, and geopolitical risk data



Multi-Agent Debate

Parallel agent reasoning for decision validation and confidence scoring



Automated ERP Execution

Direct booking and shipment execution through enterprise systems



Trust the Route. See the Reasoning.

AeroTrade XAI replaces blind trust with full transparency – every recommendation backed by auditable logic, exposed in real time.

Open Source

Full codebase available for audit and extension

Enterprise Ready

Designed for integration with existing logistics stacks

Built to Explain

Every decision traceable, every formula visible